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Department of Computer Science

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This is to certify that Mr./ Ms. Fuad Bin Zafar.....
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Prof. Rajiv Ganguly
Dean, University of Engineering & Management

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Certificate ID

Mobile/Online Voting System

A project report submitted for the partial fulfillment of the
Bachelor of Technology Degree in
Computer Science & Engineering under
Maulana Abul Kalam Azad University of Technology by

Fuad Bin Zafar

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CERTIFICATE

TO WHOM IT MAY CONCERN

This is to certify that the project report titled “**Mobile/Online Voting System**”, submitted by **Fuad Bin Zafar, Roll No: 10400117238, Registration Number: 171040110698, Apo Dhar, Roll No: 10400117241, Registration Number: 171040110695**, students of **Institute of Engineering & Management** in partial fulfillment of requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering**, is a bonafide work carried out under the supervision of **PROF. Ee-Kian Wong** during the final year of the academic session of 2017-2021. The content of this report has not been submitted to any other university or institute for the award of any other degree.

It is further certified that the work is entirely original and the performance has been found to be satisfactory.

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**DECLARATION
FOR NON-COMMITMENT OF PLAGIARISM**

We, Fuad Bin Zafar and Apo Dhar, students of B.Tech in the Department of Computer Science and Engineering, Institute of Engineering & Management have submitted the project report in partial fulfillment of the requirements to obtain the above noted degree. We declare that we have not committed plagiarism in any form or violated copyright while writing the report and have acknowledged the sources and/or the credit of other authors wherever applicable. If subsequently it is found that we have committed plagiarism or violated copyright, then the authority has full right to cancel/reject/revoke our degree.

Name of the Student: Fuad Bin Zafar

Full Signature: _____

Date: _____

Name of the Student: Apo Dhar

Full Signature: _____

Date: _____

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EXECUTIVE SUMMARY

Over the last few years, there have been a number of election observers who have suggested introducing electronic voting election processes. A general observation is that as more business is done using electronic mediums, it should not be difficult to carry out voting using electronic equipment rather than turning up at the polling place on voting day to use paper and pen. The Online Voting System (OVS) under implementation mainly addresses the voting phase. Electronic voting using the OVS should be cheaper than the present paper based arrangement. The phenomenal use of the Internet as a vehicle for improving communication, access to information and electronic commerce has led to the claim that the Internet could be used as either a replacement to attendance voting or as an additional voting option. Throughout history, election fraud has occurred in many electoral processes from which experience shows that the manual voting process is a major source of such vices and violence in many Organizations.

The mechanism leading to fraud is manifested in registration places by corrupt officials who are in a position to issue voter registration data capture forms to illegitimate individuals, stuff ballot boxes, invalidate registration for opposition voters or even coerce voters. Therefore OVS shall reduce the time spend making long queues at the polling stations during voting. It shall also enable the voters to vote from any branch of the globe since this is an online application available on the internet. Cases of vote miscounts shall also be solved since at the backend of this system resides a well developed database using MYSQL that can provide the correct data once it's correctly queried. Since the voting process shall be open as early as possible, the voters shall have ample time to decide when and whom to vote for.

The main aspect behind OVS is that it enabled us to bring out the new ideas that were sustained within us for many for many days. This project offers the voters to cast easily through internet. Vote counting is also made easy by the OVS since it's just a matter of querying the database. OVS is used by a number of countries today. Developing a good system is critical to the success of the system to prevent system failures and to gain wide acceptance as the best method available. OVS will be an inexpensive and less time consuming method once a system exhibited. OVS serves to be the best In use especially in the 21st century where human beings are embracing technology and where there is malicious struggle for power by leaders all over the world. This struggle for power has resulted in the use of all approaches by the leaders in power to remain in their positions at whatever costs even if it means applying vote rigging to win elections. With this system in place, a number of such problems shall be forgotten. I therefore hope that every organization should put the OVS technology at practice to phase out some of the problems they go through during manual voting.

CHAPTER I: INTRODUCTION

1.1 Background

The Online voting system (OVS) also known as e-voting is a term encompassing several different types of voting embracing both electronic means of counting votes. Electronic voting technology can include punched cards, optical scan voting systems and specialized voting kiosks (including self contained direct-recording electronic voting systems or DRE). It can also involve transmission of ballots and votes via telephones, private computer networks, or the internet.

Online voting is an electronic way of choosing leaders via a web driven application. The advantage of online voting over the common “queue method” is that the voters have the choice of voting at their own free time and there is reduced congestion. It also minimizes on errors of vote counting. The individual votes are submitted in a database which can be queried to find out who of the aspirants for a given post has the highest number of votes.

This system is geared towards increasing the voting percentage in Nepal since it has been noted that with the old voting method {the Queue System}, the voter turnout has been a wanting case. With system in place also, if high security is applied, cases of false votes shall be reduced.

With the “ONLINE VOTING SYSTEM”, a voter can use his\her voting right online without any difficulty. He\She has to register as a voter first before being authorized to vote. The registration should be done prior to the voting date to enable data update in the database.

However, not just anybody can vote. For one to participate in the elections, he/she must have the requirements. For instance, he/she must be a registered citizen i.e. must be 18 and above years old. As already stated, the project ‘Online Voting’ provides means for fast and convenient voting and access to this system is limited only to registered voters.

Internet voting systems are appealing for several reasons which include; People are getting more used to work with computers to do all sorts of things, namely sensitive operations such as shopping and home banking and they allow people to vote far from where they usually live, helping to reduce absenteeism rate.

1.2 Objectives of the Project

The specific objectives of the project include:

- ❖ Reviewing the existing/current voting process or approach in Organization ;
- ❖ Coming up with an automated voting system in Organization;
- ❖ Implementing a an automated/online voting system;
- ❖ Validating the system to ensure that only legible voters are allowed to vote.

1.3 Scope of the Project

It is focused on studying the existing system of voting in and to make sure that the peoples vote is counts, for fairness in the elective positions. This is also will produce:

- Less effort and less labor intensive, as the primary cost and focus primary on creating, managing, and running a secure web voting portal.
- Increasing number of voters as individuals will find it easier and more convenient to vote, especially those who are abroad having name on voter list

1.4 Project Framework

A framework is a standardized set of concepts, practices, and criteria for dealing with a common type of problem, which can be used as a reference to help us approach and resolve new problems of a similar nature.

The aim of framework is to provide a common structure so that developers don't have to redo it from scratch and can reuse the code provided. In this way, frameworks allows us to cut out much of the work and save a lot of time.

1.5 Limitation

Due to covid pandemic situation we not able to buy the Mantra MFS100 Fingerprint Sensor. In case of the Mantra MFS100 Fingerprint Sensor we used the CloudABIS biometric Fingerprint .Which is basically work our android fingerprint sensor.

CHAPTER II: BACKGROUND STUDY

2.1 Background survey

JavaScript is a high-level, dynamic, programming language that is one of the most popular and widely used programming languages in the world today as most of the websites and web browsers use and support JavaScript. With JavaScript becoming very popular, many tools, libraries, and frameworks are built upon it over the years, that solves the common programming problems more easily and efficiently. Often the lifespan of the frameworks/libraries is not too long as a newer one gets to be introduced almost every 6 months claiming to have a better user interface development according to another study published on stack overflow website. With the rapid changes in the tech industry regarding the frameworks and libraries, developers are forced to keep up and constantly learn new technologies which are commonly referred to as JavaScript Fatigue. Each of these technologies come with their advantages and disadvantages which makes the process of decision making when building a web application pretty hectic.

2.2 Fingerprint Sensor

Fingerprint Module consists of optical fingerprint sensor, high-speed DSP processor, high performance fingerprint alignment algorithm, high-capacity FLASH chips and other hardware and software composition, stable performance, simple structure, with fingerprint entry, image processing, fingerprint matching, search and template storage and other functions. Fingerprint module has two interface TTL UART and USB2.0, USB2.0 interface are often connected to the computer; RS232 interface may be a TTL level, the default baud is 57600, can be changed, ask a communication protocol, microcontroller, like ARM, DSP and other serial devices with a connection, 3.3V- 5V microcontroller are often connected directly.



Figure 1: Mantra MFS100 Fingerprint Sensor

Mantra MFS100 is a high-quality aadhaar enabled fingerprint scanner based on optical sensing technology that can be used for multiple applications such as aadhaar verification, identification, and time-attendance. This device can be extensively used for government-run programs and social welfare schemes such as Jeevan Pramaan, AePS, NDLM, CSCs to name a few. This fingerprint scanner is safe and easy to use biometric device having superior execution, precision, and toughness. MFS100 is an STQC certified product and has a plug and play USB 2.0 high-speed interface that supports multiple devices handling.

2.3 Specification

❖ In The Box

- 1 Mantra Device, 1 User manual

❖ General

- Brand MANTRA
- Model Number Mfs 100 With RD
- Backup Options yes
- LCD Yes
- LCD Resolution 500 DPI
- Scratch Proof Yes
- Multiple Languages No
- Material Plastic
- Identification Method Fingerprint
- Fingerprint Capacity 100000000
- Card Capacity 0
- Power Requirement USB
- Identification Speed < 0.8 sec.
- Type Door Locks, Time & Attendance, Payment Device, Access Control
- Model Name MFS 100
- Suitable For Computer, Laptop, Android phone

❖ Additional Features

- Alarm ☐ No
- Waterproof ☐ No
- Certification ☐ Yes
- ☐

❖ **Dimensions**

- Width 16 cm
- Height 11 cm
- Depth 6 cm
- Weight 0.2 kg

2.4 How work the fingerprint Biometric

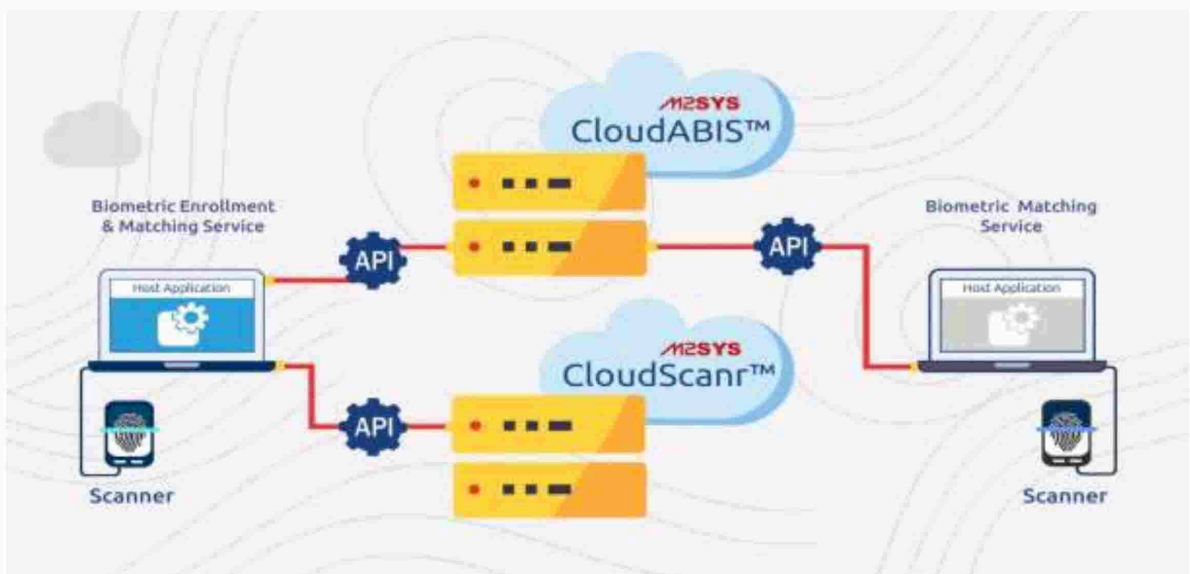


Figure 2: CloudABIS biometric Fingerprint working procedure

The Following simple steps below to learn how to configure CloudABIS biometric API with your web or mobile application:

❖ **Step-01: Subscribe to CloudABIS**

To avail the service you need to choose a subscription plan according to your project or business requirement. The SaaS (Software-as-a-Service) based pricing model provides the flexibility to

scale-up or scale-down whenever you need it. After the successful subscription, you will get biometric credentials for the followings to configure your web or mobile applications:

- API URL
- App Key
- Secret Key
- Customer Key
- Engine Name
- (Keep the credentials confidential for the security of your system and further use.)

❖ **Step-02: API Authentication**

In this step, you need to log-in to the admin panel of your web or mobile application. Following the video instruction or as your convenient configure the API URL, App Key, and Secret Key in your code editor. The API will be authenticated as the output and will be ready to use.

❖ **Step-03: Configuring Biometric**

After API authentication you will be allowed to use biometric but you need to configure it for the first use. Configure the biometric identification using Customer Key, and Engine Name to register, identify, and verify the users. To activate the biometric capture tool you also need to follow Biometric XML or you can avail a ready biometric capture tool named CloudScanr. The technical configuration process is done already. Now you can register the users creating a user ID and capturing their biometric data using your application.

❖ **Step-04: Enrollment, Identification, and Verification**

You are all set now to identify and verify the users from your web or mobile application. Enroll the users registering their user ID and biometrics to identify or verify them. CloudABIS provides the faster biometric matching and can perform millions of matching within a second.

❖ **Step-05: Update of User Information**

A system may require the update of user information and CloudABIS allows the system administrator to update biometric data and user ID of the users to smoothen the biometric identification system.

❖ **Step-06: Checking Existing User**

Using “*isRegister*” condition, CloudABIS enables the operator to check if the user is already registered in the system or not. It eases the work process of the system admin or operator and helps to eliminate the chances of duplicate user enrollment into the system and ensures unbiased user validation.

2.5 Security Issues of Online Voting

Foreign experience revealed that they are often confronted by security issues while the online voting system is running. The origin of the security issues was due to not only outsider (such as voters and attackers) but also insider (such as system developers and administrators), even just because the inheritance of some objects in the source code are unsuitable. These errors caused the voting system to crash.

The proposed solutions were correspondingly outlined to hold back these attacks. For example, to avoid hacker making incursion into the voting system via network, we can design our system to transmit data without network. Another example is to limit voter to input particular data, so that we can prevent the command injection from running.

CHAPTER III: REQUIREMENTS

3.1 Tools Used

❖ **Xampp:**

○ **Apache:**

- (Application Server) Apache , often referred to as Server, is an open-source Java Servlet Container developed by the Apache Software Foundation.

○ **MySQL Server:**

- It handles large databases much faster than existing solutions.
- It consists of multi-threaded SQL server that supports different back ends, several different client programs and libraries, administrative tools, and application programming interfaces (APIs)
- Its connectivity, speed, and security make MySQL Server highly suited for accessing databases on the Internet.

❖ **Sublime Text 3.1.1-** Sublime Text is a sophisticated text editor for code, markup and prose. You'll love the slick user interface, extraordinary features and amazing performance.

❖ **Web browsers:** Google Chrome, Mozilla Firefox, Opera and Internet Explorer.

❖ **Git Hub:** GitHub Inc. is a web-based hosting service for version control using Git. It is mostly used for computer code. It offers all of the distributed version control and source code management functionality of Git as well as adding its own features.

❖ **Android Studio:** It is an opensource Android application. It is used for implementing biometric authentication.

3.2 Functional Requirements

3.2.1 Use case Diagram

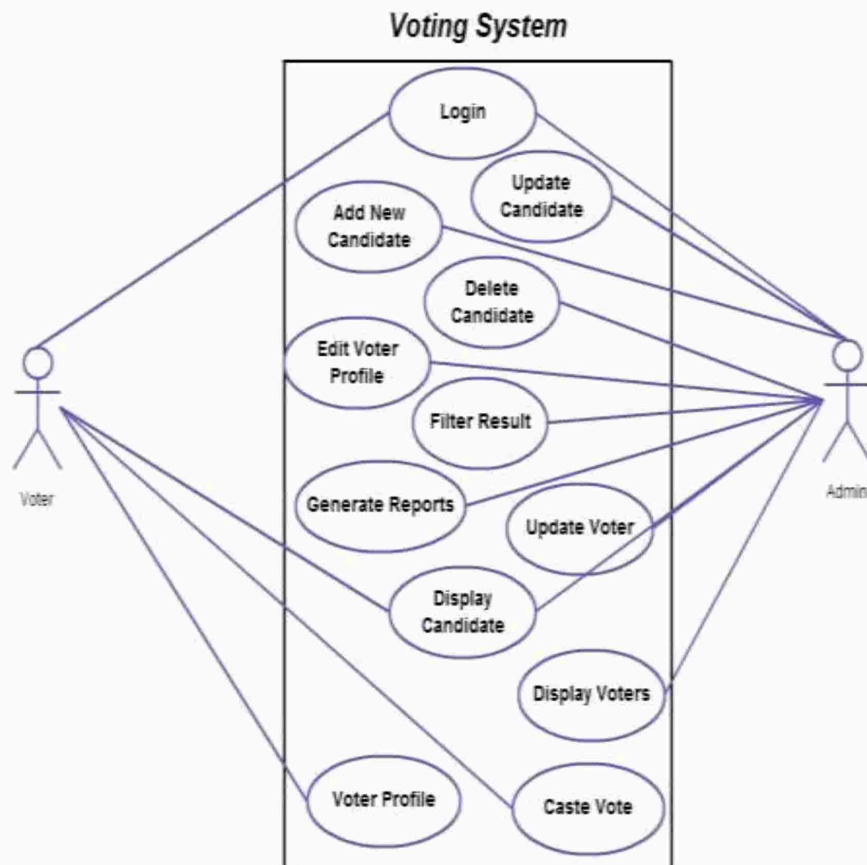


Figure 3: Use case Diagram

Above figure represents Use Case Diagram of the project and is a useful technique for identifying, clarifying, and organizing system requirements. It describes how a user uses a system to accomplish a particular goal. Use cases help ensure that the correct system is developed by capturing the requirements from the user's point of view.

3.2.2 Admin Use case Description

Use Case ID:	08		
Use Case Name:	Admin Dashboard		
Created By:	Fuad Bin Zafar	Last Updated By:	Fuad Bin Zafar
Date Created:	06/07/2021	Date Last Updated:	06/07/2021

Actor:	Admin
Description:	In this Option, the Admin can create his/her account to provide various services to the user. The admin add voters details, candidate details, candidate position, election title, create result and view result.
Preconditions:	Need to have a proper user id and password for login the portal for maintaining the information.

Postconditions:	Admin can update voter list, candidate list, election details etc and database will also update.
Frequency of Use:	Multiple times
Flow of Events:	System: Shows the dashboard of the voting system. User: View the details of the records.

Alternative Flows:	1) <u>Change user name and password</u> System: Shows the details of the user with a Change the user name and Password option. User: Clicks on the option. System: Asks for the old password and new password. User: Gives the old password and new Password. System: Confirms the change of the password.
----------------------------------	--

3.2.3 Voter Use case Description

Use Case ID:	02
Use Case Name:	Voters cast vote

Created By:	Fuad Bin Zafar	Last Updated By:	Fuad Bin Zafar
Date Created:	06/07/2021	Date Last Updated:	06/07/2021

Actor:	Voter
Description:	In this Option, the voter verify his finger print for entering the web portal and login for cast vote.
Preconditions:	Need to have a proper user id and password for login the portal for cast vote.
Postconditions:	Voter cast his vote to his choice able candidate.
Frequency of Use:	One time

Flow of Events:	System: Shows the candidate list. User: Cast his vote.
------------------------	---

3.2.4 Sequence Diagram

A sequence diagram is a type of interaction diagram because it describes how—and in what order—a group of objects works together. A sequence diagram specifically focuses on lifelines, or the processes and objects that live simultaneously, and the messages exchanged between them to perform a function before the lifeline ends.

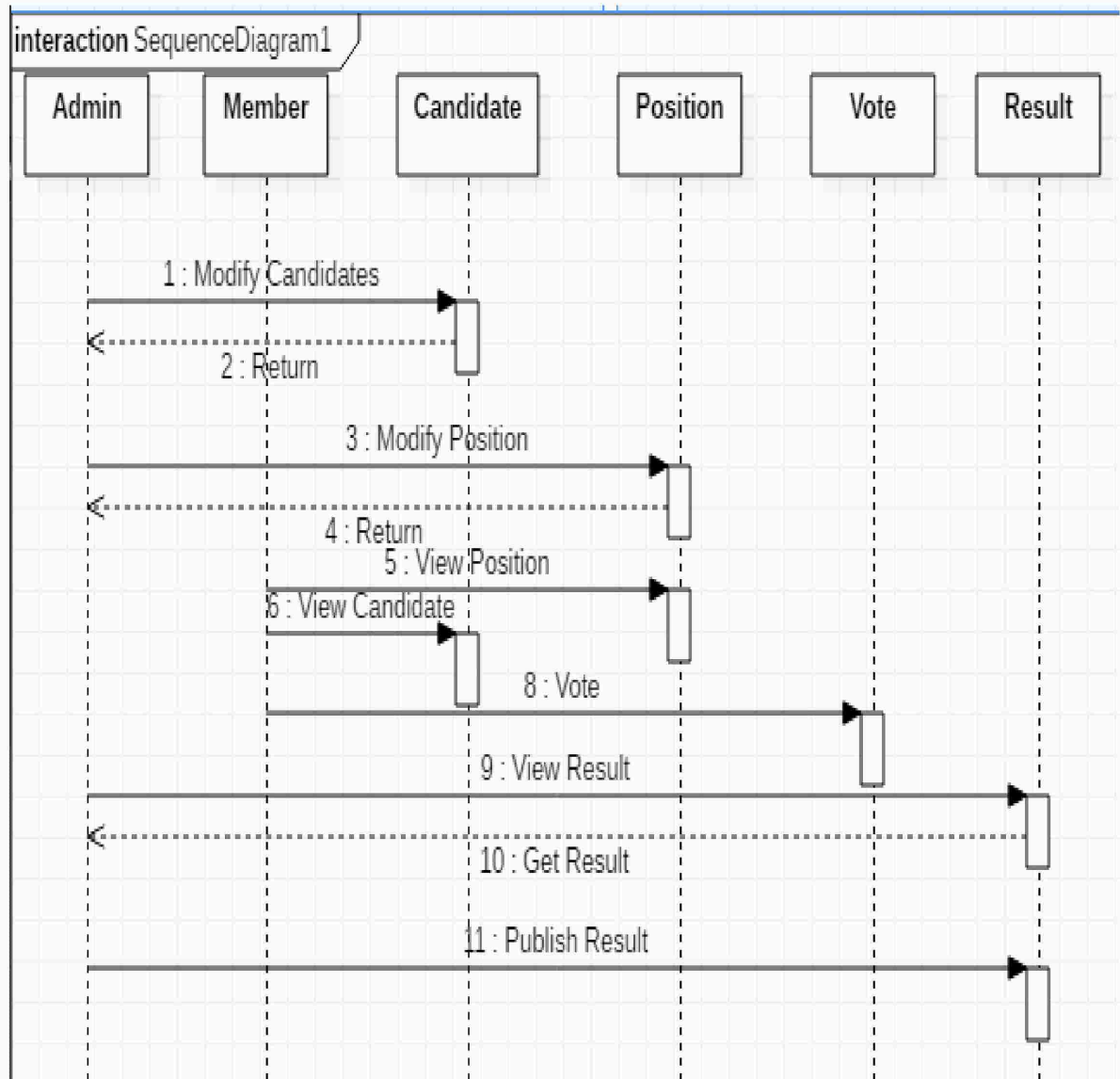


Figure 4: Sequence Diagram

Above diagram represents Sequence Diagram of the project which is a type of interaction diagram because it describes how—and in what order—a group of objects works together. A sequence diagram specifically focuses on lifelines, or the processes and objects that live simultaneously, and the messages exchanged between them to perform a function before the lifeline ends.

3.2.5 Data Flow Diagram

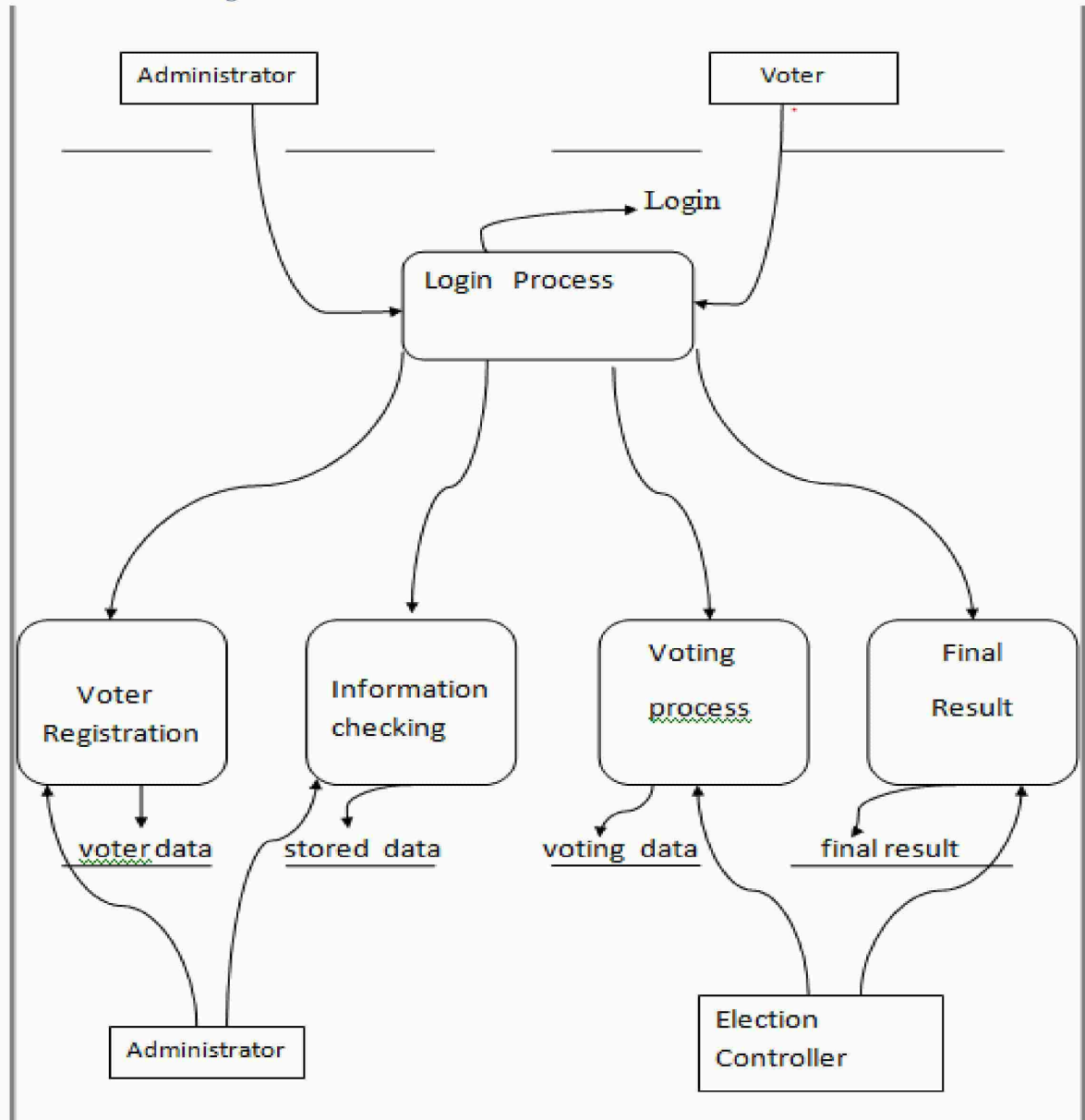


Figure 5: Dataflow Diagram

Above Data Flow Diagram, explains the overall structure of the system. It shows how and what types of services the client chooses and the amount of admin interaction in it.

3.2.6 Activity Diagram

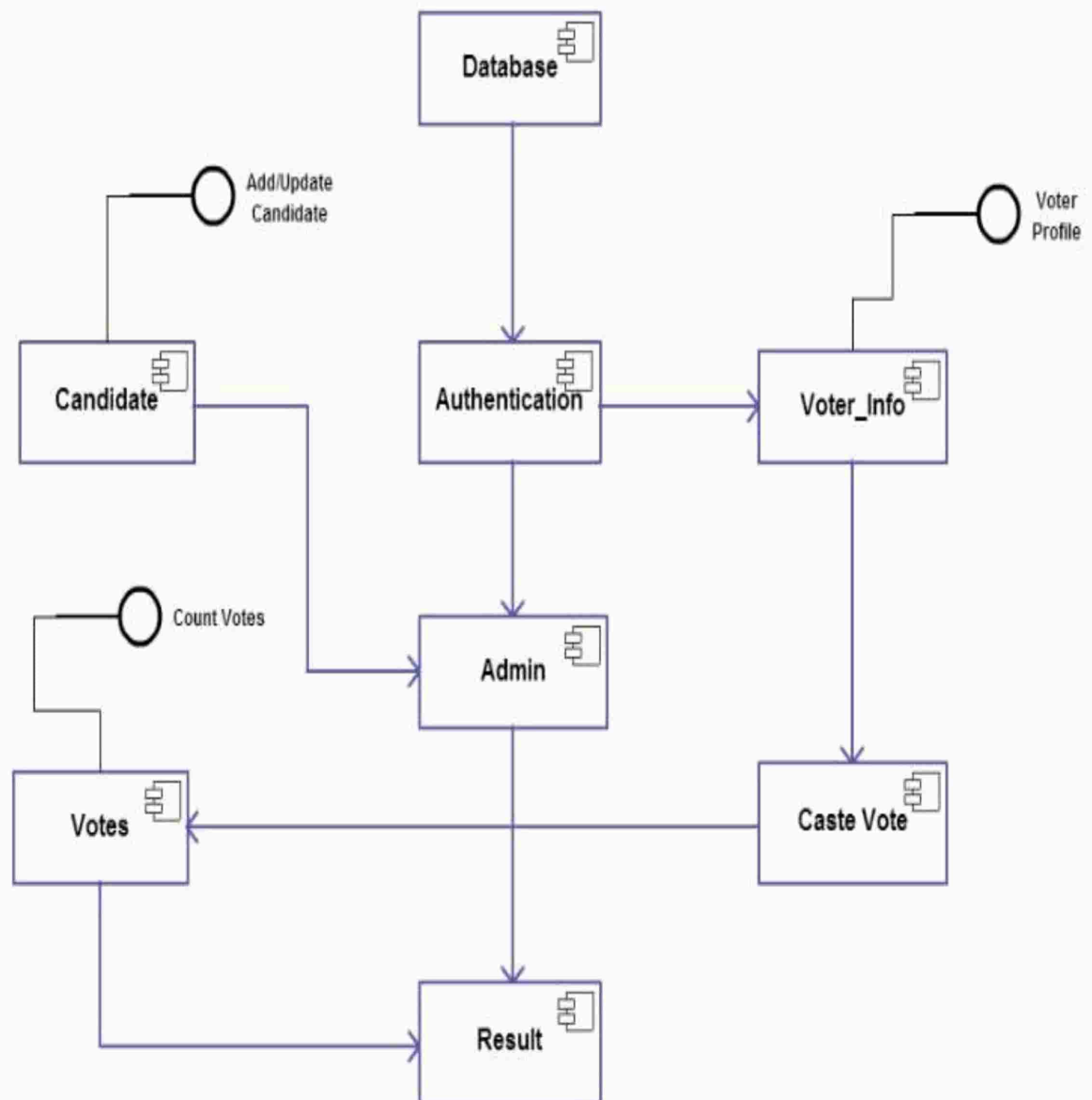


Figure 6: Activity Diagram

3.3 Non-Functional Requirements:

3.3.1 Security:

In future, we will make our website more secure by enabling more functionalities like sql injection so that hackers would never get any confidential information of voter system. Moreover, we Will increase the security so that any possible change in vote count will never happen.

CHAPTER IV: DESIGN

4.1 Admin Table

Id	Username	Password	First name	Last Name	Photo	Created Date
1	Code Builder	Fuad1234	Code	Builder	Avatar.png	20/4/2021

4.2 Candidate Table

Id	Position Id/Region	First Name	Last Name	Photo	Platform
30	12	Agnik	Saha	Agni.jpg	2 no Ward
31	13	Fuad	Bin Zafar	Fuad.jpg	1 no ward

4.3 Voter Table

Id	Voter's Id	Password	First Name	Last Name	Photo
30	1003	12345	Apo	Dhar	Apo.jpg
31	1001	123	Agnik	Saha	Agnik.jpg
32	1002	1234	Fuad	Bin Zafar	Fuad.jpg

4.4 Position Table

Id	Description	Max_Vote	Priority
12	President	7	1

4.4 Vote's table

Id	Voter's_Id	Candidate Id	Position Id/Region
34	1003	30	12
35	1001	30	12
36	1001	31	13

CHAPTER V: IMPLEMENTATION

5.1 Java script input Validation

It is important to validate the form submitted by the user because it can have inappropriate values. So, validation is must to authenticate user.

JavaScript provides facility to validate the form on the client-side so data processing will be faster than server-side validation. Most of the web developers prefer JavaScript form validation.

Through JavaScript, we can validate name, password, email, date, mobile numbers and more fields.

In this example, we are going to validate the name and password. The name can't be empty and password can't be less than 6 characters long.

Here, we are validating the form on form submit. The user will not be forwarded to the next page until given values are correct.

```
<script>
function validateform(){
var name=document.myform.name.value;
var password=document.myform.password.value;

if (name==null || name==""){
    alert("Name can't be blank");
    return false;
}else if(password.length<6){
    alert("Password must be at least 6 characters long.");
    return false;
}
}
</script>
<body>
<form name="myform" method="post" action="abc.jsp" onsubmit="return validateform()"
">
Name: <input type="text" name="name"><br/>
Password: <input type="password" name="password"><br/>
```

`<input type="submit" value="register">`

`</form>`

```
<?php
include 'includes/session.php';

if(isset($_POST['add'])){
    $firstname = $_POST['firstname'];
    $lastname = $_POST['lastname'];
    $position = $_POST['position'];
    $platform = $_POST['platform'];
    $filename = $_FILES['photo']['name'];
    if(!empty($filename)){
        move_uploaded_file($_FILES['photo']['tmp_name'], '../images/'.$filename);
    }

    $sql = "INSERT INTO candidates (position_id, firstname, lastname, photo, platform) VALUES ('
$position', '$firstname', '$lastname', '$filename', '$platform')";
    if($conn->query($sql)){
        $_SESSION['success'] = 'Candidate added successfully';
    }
    else{
        $_SESSION['error'] = $conn->error;
    }
}
else{
    $_SESSION['error'] = 'Fill up add form first';
}

header('location: candidates.php');
?>
```

5.2 Database Connection

```
<?php
$conn = new mysqli('localhost', 'root', '', 'votesystem');

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
```

?>

5.3 Application's Output

5.4 Admin Part

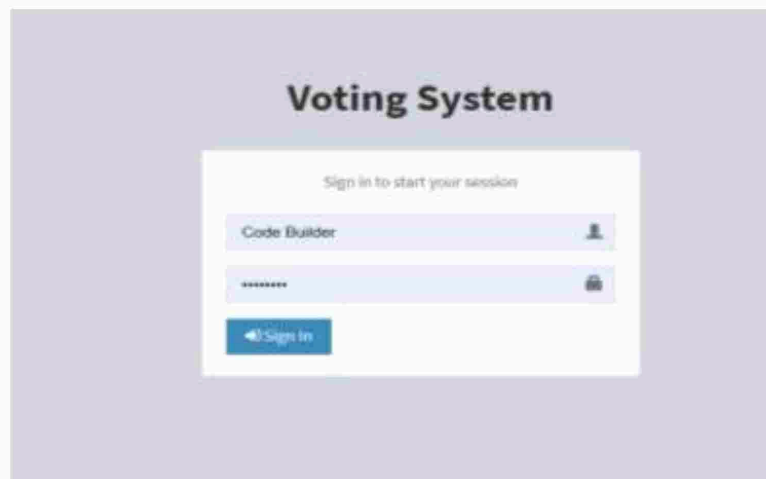


Figure 7: Admin/Member Login

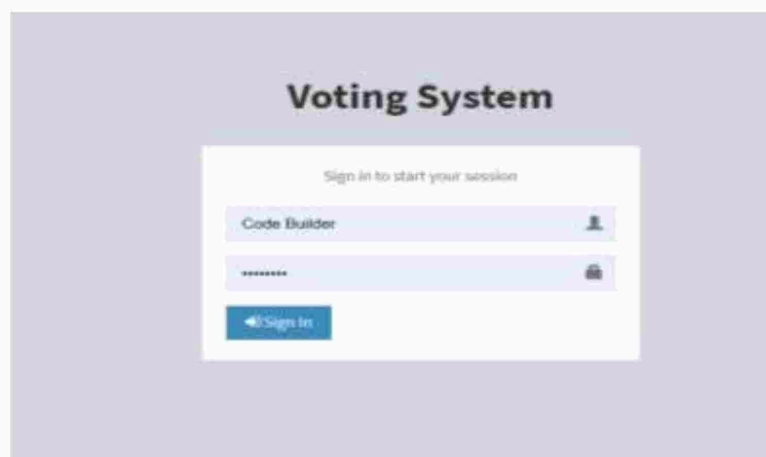


Figure 8: Admin Login

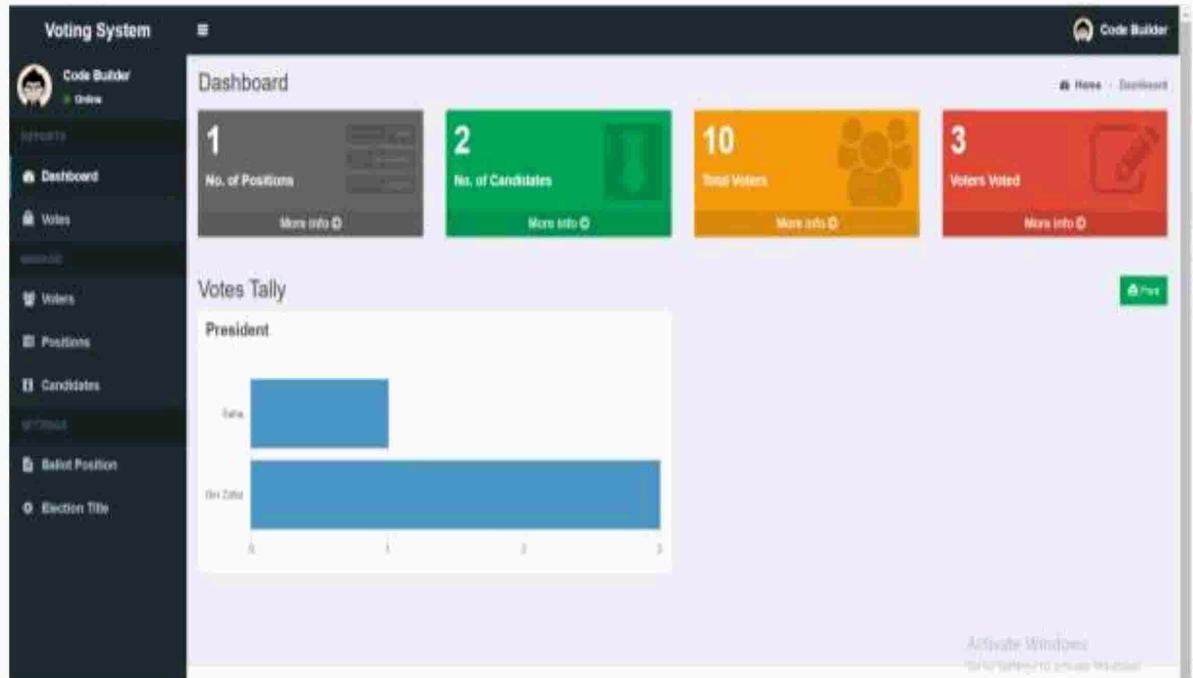


Figure 9: Dashboard

Voting System View Votes

Position	Candidates	Voter
President	Pooji Bin Zafar	Agnik Sara
President	Pooji Bin Zafar	Anousha Sara
President	Pooji Bin Zafar	Apo Dhar
President	Agnik Sara	Apo Dhar

Showing 1 to 4 of 4 entries

Figure 10: View Votes

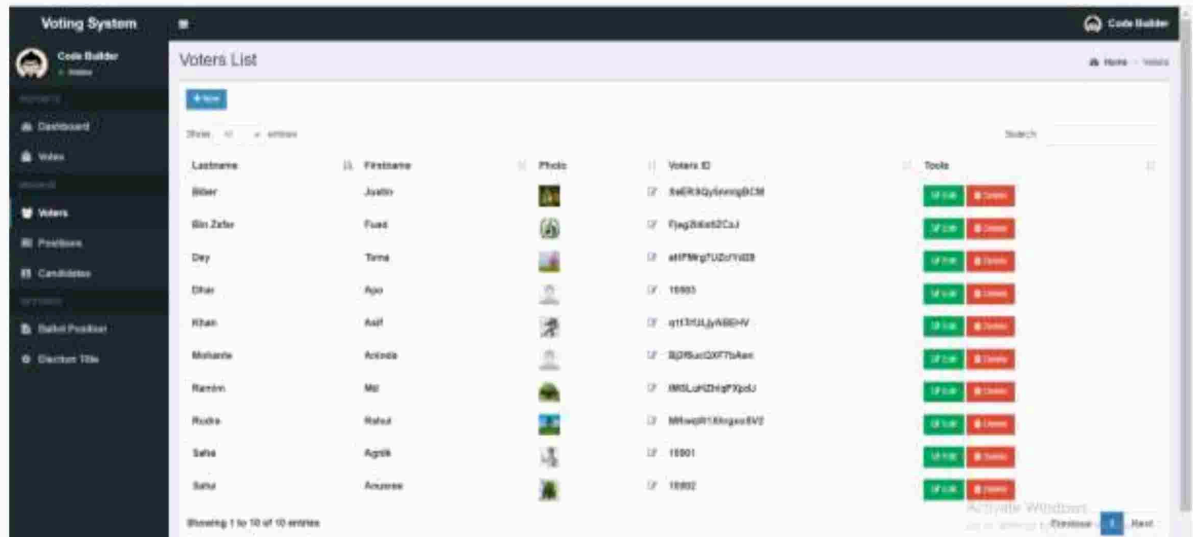


Figure 11: Manage Voters

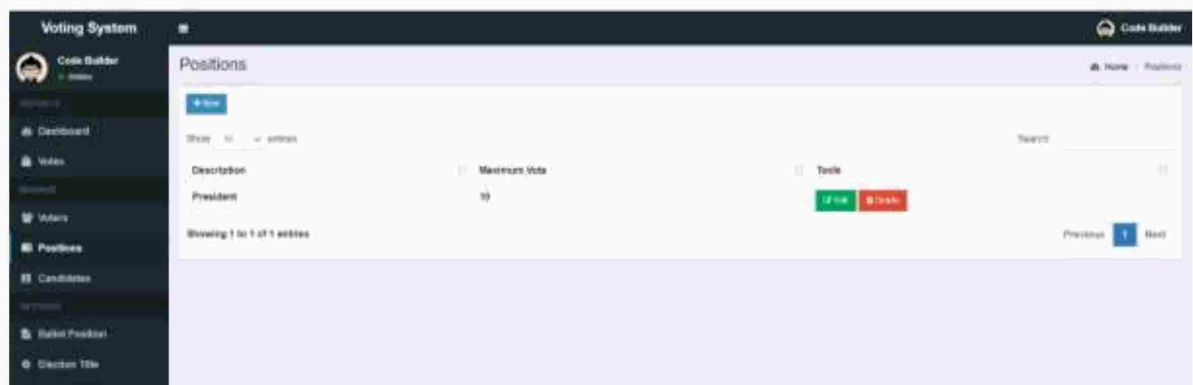


Figure 12: Manage Position

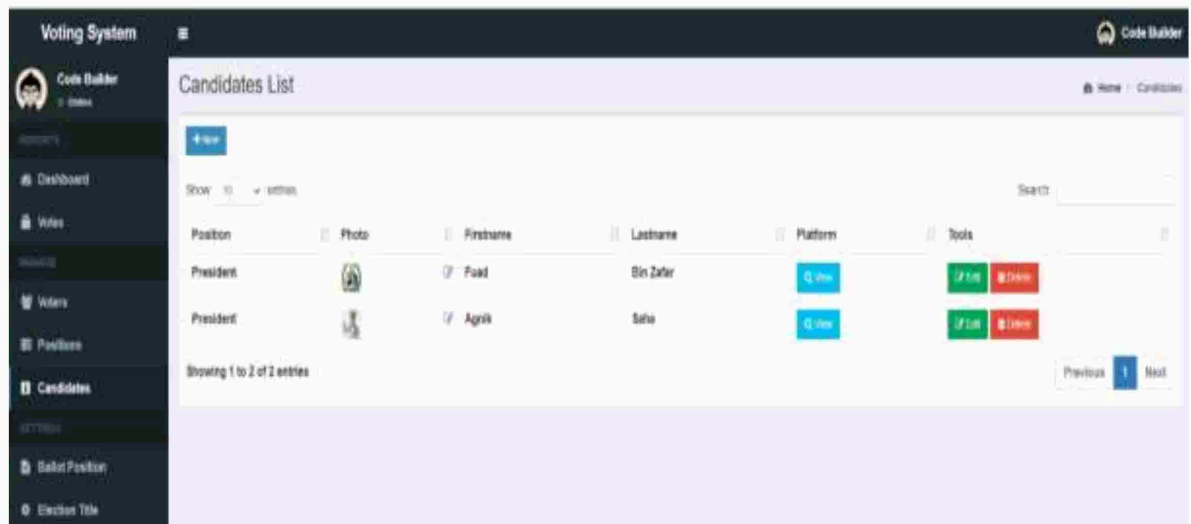


Figure 13: Manage Candidates

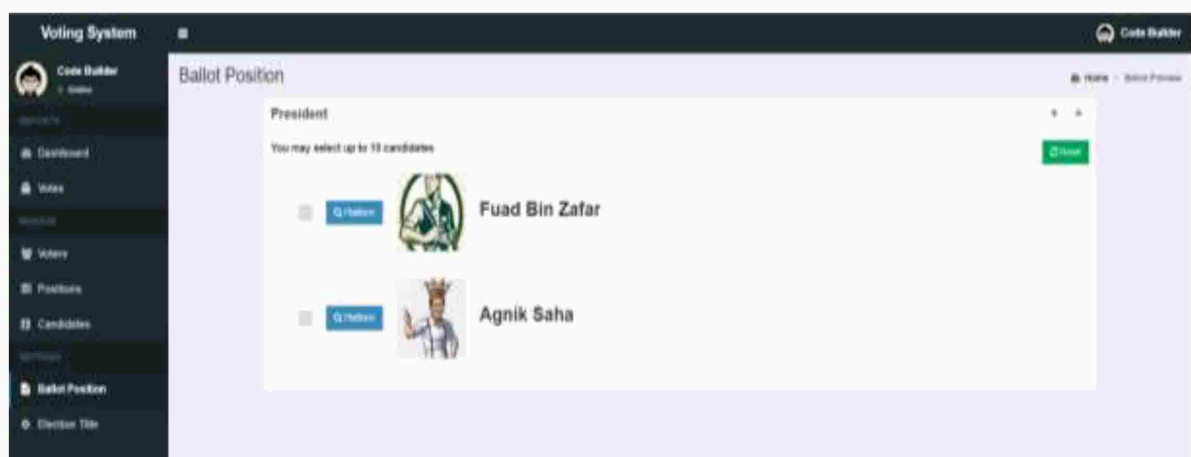


Figure 14: Manage Ballot Position

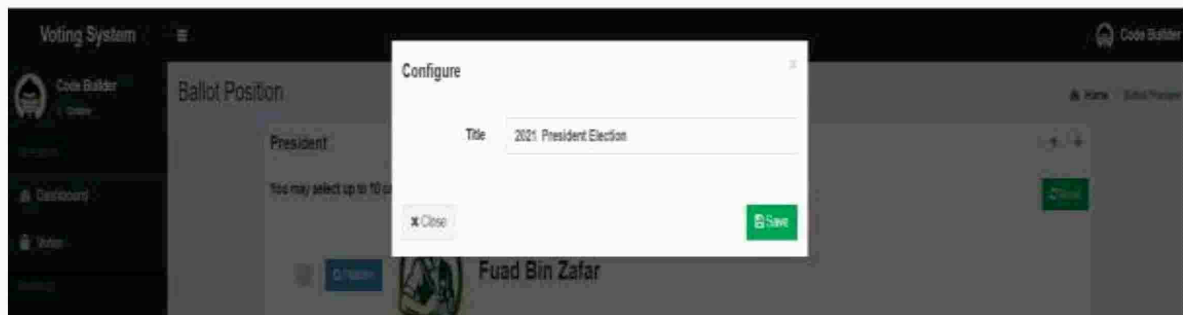


Figure 15: Election Title

5.5 Voter's Part

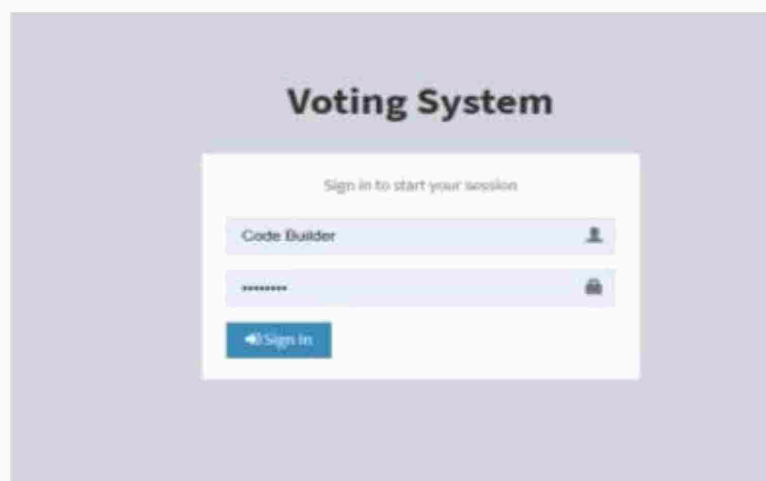


Figure 16: Voter's Login

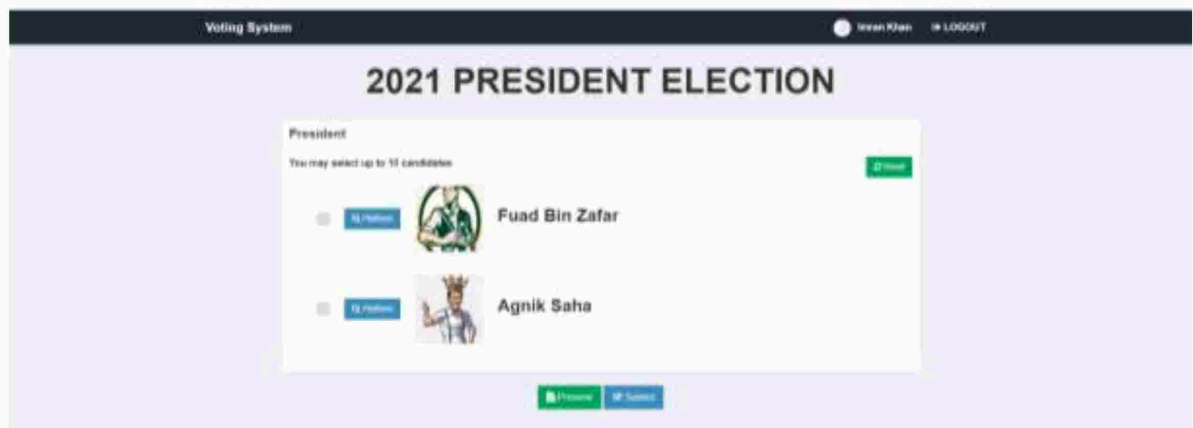


Figure 17: Cast Vote

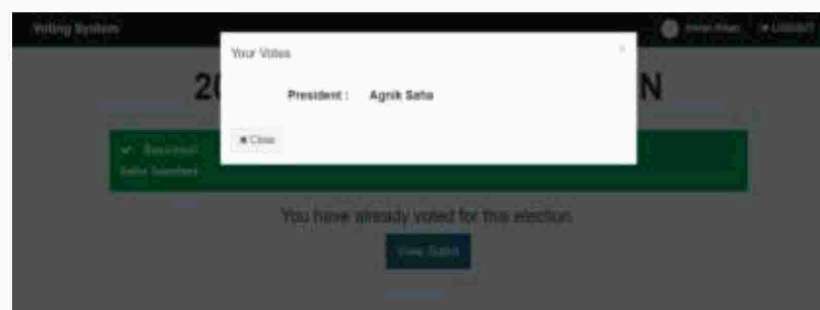
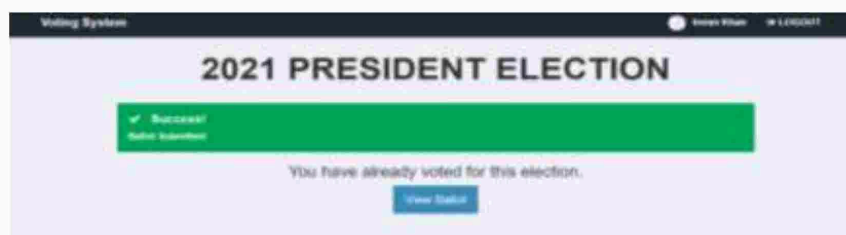


Figure 18: View Votes and Logout

5.6 Biometric Authentication

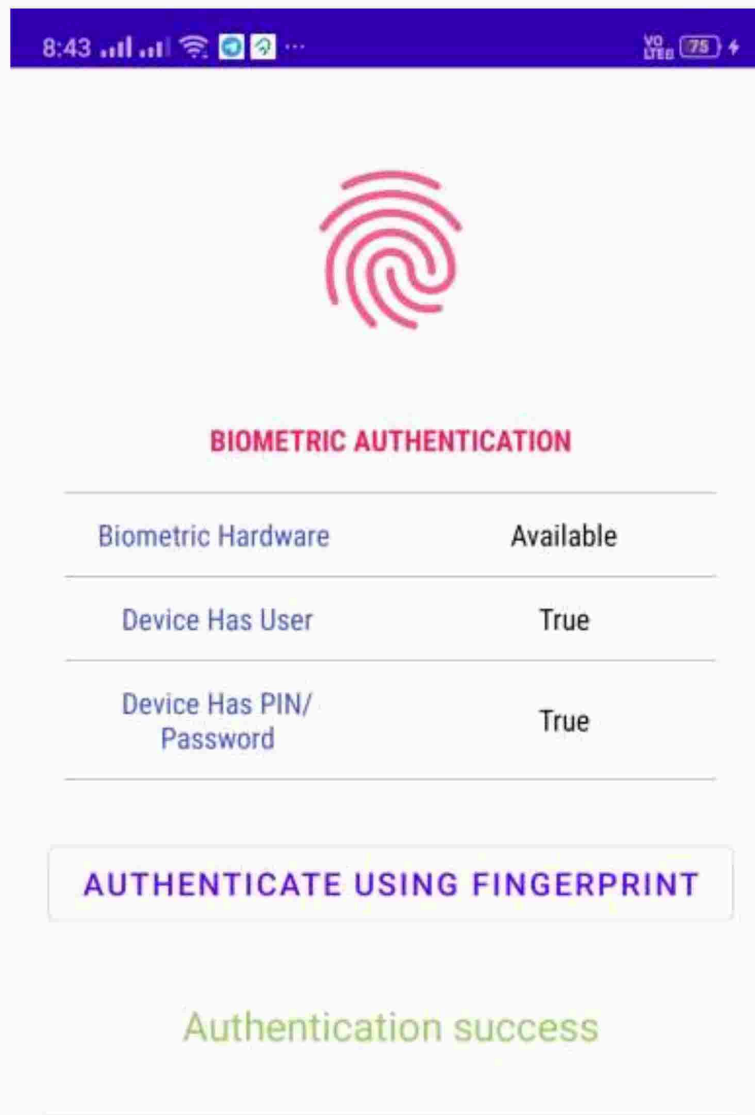


Figure 19: Biometric Authentication

CHAPTER VI: TESTING

6.1 Test Case

Test Scenario ID	Test Scenario Description	Test Case ID	Test Case Description
TCS_PRS_001	Verify the login functionality	TC_PRS_Login_001	Enter a valid User id & valid password
TCS_PRS_002	Verify the login functionality	TC_PRS_Login_002	Enter a valid User Id & invalid password
TCS_PRS_003	Verify the login functionality	TC_PRS_Login_003	Enter an invalid User Id & valid password
TCS_PRS_004	Verify the login functionality	TC_PRS_Login_004	Enter an invalid User I'd & invalid password
TCS_PRS_005	Verify the login functionality	TC_PRS_Login_005	Enter none of the credentials
TCS_PRS_006	Verify the login functionality	TC_PRS_Login_006	Enter only password
TCS_PRS_007	Verify the login functionality	TC_PRS_Login_007	Enter only User Id

Test steps	Pre Conditions	Test data	Post Conditions
1.Enter valid User id 2. Enter valid password 3. Click login button	Valid URL Test Data	User id : Code Builder Password: Fuad1234	Redirect to dashboard page
1.Enter User id 2. Enter invalid password 3. Click login button	Valid URL Test Data	User id : Code Builder Password: Fuad1233	Error: Invalid User id or Password
1.Enter User id 2. Enter valid password 3. Click login button	Valid URL Test Data	User id : Code Builder.1 Password: Fuad1234	Error: Invalid User id or Password
1.Enter User id 2. Enter invalid password 3. Click login button	Valid URL Test Data	User id : Code Builder Password: Fuad1235	Error: Invalid User id or Password
1. Click login button	Valid URL Test Data	User id : Code Builder Password: Fuad1234	Redirect to dashboard page
1.Enter password 2.Click Login	Valid URL Test Data	User id : Code Builder Password: Fuad1234	Please fill out this field
1.Enter User id 2.Click Login	Valid URL Test Data	User id : Code Builder Password: Fuad1234	Please fill out this field

Expected Results	Actual Results	Status
Login Successful	Login Successful	Pass
Error: Invalid Email or Password	Login Successful	Fail
Error: Invalid Email or Password	Error: Invalid Email or Password	Pass

Error: Invalid Email or Password	Error: Invalid Email or Password	Pass
Message: Please fill out this field	Login Successful	Fail
Message: Please fill out this field	Error: Please fill out this field	Pass
Message: Please fill out this field	Error: Please fill out this field	Pass

A test case is a specification of the inputs, execution conditions, testing procedure, and expected results that define a single test to be executed to achieve a particular software testing objective, such as to exercise a particular program path or to verify compliance with a specific requirement.

CHAPTER VII: CONCLUSION & DISCUSSION

7.1 Conclusion

This Online Voting system will manage the Voter's information by which voter can login and use his voting rights. The system will incorporate all features of voting system. It provides the tools for maintaining voter's vote to every party and it count total no. of votes of every party. There is a database which is maintained by the Sarwamangal Youth Club in which all the names of voter with complete information is stored.

In this member who had registered his/her information on the database and when he/she want to vote he/she has to login by his email and password and can vote to any candidate only single time. Voting detail store in database and the result is displayed by calculation. By online voting system percentage of voting is increases. It decreases the cost and time of voting process. It is very easy to use and it is vary less time consuming. It is very easy to debug.

7.2 Future Enhancement

With the existing constraints, the developed systems is not what was planned initially. The primary aim of this project has been met. All the objectives that were set out have been completed and giving positive results in the ends. In the future some features that can be added will be about the two factor authentication. Although the user requirements were successfully met the application is not yet fully utilized because the users of this website are just learning about the benefits and working of the website. The user testing and evaluation of the application did however highlight rooms for the expansion. The application could therefore be developed further as soon as the user is fully aware of its working.

REFERENCE

- <https://www.w3schools.com/php/default.asp>
- <http://www.tizag.com/cssT/> (for css coding reference)
- <http://www.tizag.com/mysqlTutorial/> (for the use of tables)